Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/17/2007
PI Name:	Wilson, John W. Ph.D.		
Project Title:	Consortium for Shielding Models and Computational Procedures		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	757-864-1414
Organization Name:	NASA Langley Research Center		
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City:	Hampton	**State:**	VA
Zip Code:	23681-0001	**Congressional District:**	1
Comments:	srhp.jsc.nasa.gov		
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-05
Start Date:	11/15/2002	**End Date:**	11/14/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tripathi, Ram (NASA Langley Research Center) Heinbockel, John (Old Dominion University) Tweed, John (Old Dominion University)		
Grant/Contract No.:	Intramural		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To systematically treat physical perturbation approach as first order reduction of solutions in terms of continuous slowing down plus straggling followed by treatment of multiple elastic scattering and nuclear reaction. The nuclear reaction portion is reduced to forward and isotropic component. Multiple elastic scattering and the forward nuclear reactive component are treated in asymptotic expansion about forward direction. Two approaches are used to solve forward problem: perturbation expansion and marching procedures. The first three perturbation terms are solved analytically and higher order terms evaluated using non-perturbative methods and used to renormalize third perturbation term to account to all orders. The second asymptotic term is then evaluated as off axis correction. The isotropic source of particles from nuclear reactions will be evaluated by first order perturbation theory and a new isotropic source evaluated and consist mainly of relatively low energy neutrons and are treated by approaches similar to those used in the nuclear power community. A current version of the code will be used to study the sensitivity to nuclear fragmentation cross section error. To develop a high performance 3D Boltzmann equation solver for space and laboratory use and investigate sensitivity of solutions to uncertainties in the nuclear cross sections. Protecting the astronaut from the hazards of space radiation is one of NASA's two highest priorities (O'Keefe, 2002). It is a tall pole in NASA exploration program. This task provides the software used in protecting astronauts in NASA missions and used to augment the ISS shield. Software under this program identifed by NASA for Constellation design and NCRP for astronaut safety.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	NASA identified five critical enabling technologies (roadblocks, NASA's highest development priorities) for future space exploration. "The two biggest problems are power and propulsion in deep space, and the hazardous radiation environment for humans traveling beyond Earth." http://www.spaceflightnow.com/ Astronaut health and safety are foremost considerations in future space exploration (NEXT FY02 report) and adequate provision of radiation protection is required to keep health risks at acceptable levels while maintaining acceptable mission costs. The HZETRN code is/was used in Shuttle mission safety analysis prior to each launch, Shuttle refitted avionics computers validation, ISS environment definition, ISS shield augmentation design, ISS mission safety analysis of ongoing operations (projections), astronaut radiation health assessment and organ specific exposures for astronaut medical records, RASC (Revolutionary Aerospace Systems and Concepts) studies, Mars Engineering Model definition, ESAS trade studies (Exploration Systems Architecture Study, "The 90-Day Study"), Constellation CEV design, Constellation Lunar Surface Systems design trades, Odyssey data processing, FAA aerospace studies, Living With a Star Program's atmospheric radiation studies, Space Environments and Effects Program, Space Radiation Health Project, Space Radiation Shielding Project, cross sections used in radiation therapy protocols (US, Japan, Germany),...The only code identified within the NASA design standards document STD-3000 (2005 rev) for space shield design and the only code discussed by the National Council on Radiation Protection in Operational Safety Program for Astronauts in Low-Earth Orbit: A Basic Framework. (NCRP Report No. 142, 2002). This code provides the basis for radiation protection and shielding design in future Lunar and Mars missions. This code will provide the missions safety analysis software for future NASA exploration operations. A critical enabling technology for this multi-billion dollar NASA Constellation program with high national and international visibility.
Task Progress:	To meet Constellaltion development schedules, primary attention was given to establishing/identifying a code of choice for CEV design processes now in progress. It was decided that a well established and stable older version with a history of verification and validation is preferable to a more recent research code of questionable stability. The fundamental derivations were revisited with some minor corrections in the light ion propagator and a greatly improved higher order HZE propagator with better control on the propagated error has been developed. The code was dubbed the 2005 HZETRN as it was completed in early December 2005. A new round of verification processes were undertaken with an international benchmarking activity with the Monte Carlo codes GEANT4, FLUKA, HETC, and PHITS. Only the GEANT4 and PHITS data were available at the time of the ICES manuscript preparation (see Standardized Radiation Shield Design Method: 2005 HZETRN, SAE 06ICES18). Validation processes continued to develop using the LIULIN survey on ISS 6A during June/July 2001 (see Pre-engineering validation of environmental models and the 2005 HZETRN simulation code, Adv. Sp. Res. 2007, and Time serial analysis of the induced LEO environment within the ISS 6A, Adv Sp. Res. 2007). The code was configuration controlled on 3 Mar 2006 and an Integration Working Group was established for well defined procedures in the design process. A second round of verification and validation is in progress. The Green's function version of HZETRN continues to develop with addition of Coulomb multiple scattering coupled to energy loss straggling. The comparisons with proton multiple scattering measurements are very promising. The 3D distribution of proton Bragg distributions are very encourage and detailed comparisons with experiments will take place in the near future. Coulomb multiple scattering effects on the first collision term are small and completely neglible in off axis aysmptotic expansions. The off axis components of the first collision term is in progress and will serve as the first asymptotic correction in the Green's function code. A fully 3D Green's function for space use is underdevelopment. A workshop was held at LaRC to identigy gaps and develop a plan to close them. The workshop objectives were: 1. Discuss current consortia close out 2. Communicate NASA's future needs for basic Transport capabilities (aside of coupling to engineering models) 3. Assess current status of available codes toward meeting our objectives/requirements 4. Gap Analysis for Phase II - Light ions (what is needed to meet requirements) 5. Discuss requirements for measurement and physics 6. Generate strategy document toward closing gaps The report is used by the National Academy of Science in their study, "Evaluation of Radiation Shielding for Lunar Exploration" and the PI also delivered a prospectus on design methods and materials.
Bibliography Type:	Description: (Last Updated: 08/20/2012)

Articles in Peer-reviewed Journals	Nealy JE, Cucinotta FA, Wilson JW, Badavi FF, Dacheve TsP, Tomove BT, Walker SA, De Angelis G, Blattnig SR, Atwell W. "Pre-engineering spaceflight validation of environmental models and the 2005 HZETRN simulation code." Advances in Space Research. 2007;40(11):1593–610. http://dx.doi.org/10.1016/j.asr.2006.12.029 , Oct-2007
Articles in Peer-reviewed Journals	Wilson JW, Cucinotta FA, Golightly MJ, Nealy JE, Qualls GD, Badavi FF, De Angelis G, Anderson BM, Clowdsley MS, Luetke N, Zapp N, Shavers MR, Semones E, Hunter A. "International Space Station: A testbed for experimental and computational dosimetry." Advances in Space Research. 2006;37(9):1656-63. http://dx.doi.org/10.1016/j.asr.2005.02.038 , Jul-2006
Articles in Peer-reviewed Journals	Wilson JW, Nealy JE, Dachev TsP, Tomov BT, Cucinotta FA, Badavi FF, De Angelis G, Atwell W, Leutke N. "Time serial analysis of the induced LEO environment within ISS 6A." Advances in Space Research. 2007;40(11):1562–70. http://dx.doi.org/10.1016/j.asr.2006.12.030 , Oct-2007
Articles in Peer-reviewed Journals	Mertens CJ, Wilson JW, Walker SA, Tweed J. "Coupling of multiple Coulomb scattering with energy loss and straggling in HZETRN." Advances in Space Research. 2007;40(9):1357-67. http://dx.doi.org/10.1016/j.asr.2007.03.075 , Aug-2007
Articles in Peer-reviewed Journals	Heinbockel JH, Wilson JW, Blattnig SR, Qualls GD, Badavi FF, Cucinotta FA. "HZE ion fragmentation cross-section sensitivity and propagated errors in HZE exposure estimates." Radiat Meas. 2006 Oct-Nov;41(9-10):1103-14. http://dx.doi.org/10.1016/j.radmeas.2006.02.002 , Oct-2006
Papers from Meeting Proceedings	Wilson JW. "Standardized radiation shield design methods." 2006 ICES, Norfolk, VA, July 2006. SAE ICES paper 2006-01-2109, July 2006. , Jul-2006
Patents	Provisional Application (L0045084). Patent application Dec 2006. Dec-2006 Wilson JW et al. "Standardized Radiation Shield Design Methods: 2005 HZETRN."